

Static Multipolar Order and Dynamical Chiral Response in MnF_2 : Probe–Tensor Correspondence and Conditional Rank Identifiability

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Altermagnetic MnF_2 is now described experimentally through static magnetic multipoles, strain-field thermodynamics, chiral magnons, momentum-space spin splitting, and weak symmetry-selective microscopic interactions. These descriptions can share a symmetry character without being the same physical object. We formulate a probe–tensor correspondence that separates Hamiltonian coordinates, equilibrium one-point magnetization tensors, static response tensors, connected correlations, and the probe/instrument map that produces recorded observables. The exact decomposition of the magnetic correlation tensor shows why elastic multipole reconstruction and inelastic magnon chirality cannot be identified even when they reverse under the same domain operation; for an uncontrolled domain ensemble, a vanishing ensemble one-point moment can coexist with finite diffraction intensity. We further show that symmetry selection fixes an irreducible-representation sector but need not fix multipole rank. Rank identifiability is conditional on the declared basis, radial/form-factor model, origin convention, nuisance space, and domain state. For a candidate multipole vector $\mathbf{m}_\Gamma$, the relevant criterion is the rank and conditioning of the nuisance-projected whitened kernel, not of the target block alone. As a quantitative stress test, we construct a count-matched, published-geometry surrogate for the MnF_2 polarized-neutron-diffraction experiment using the reported lattice parameters, reciprocal-space cutoff, parity rules, and symmetry-allowed basis. In the reduced even-reflection basis $\{F P_{00}, F P_{20}, \text{Mn } P_{2,-2}, \text{Mn } P_{4,-2}\}$ the normalized surrogate is well conditioned ($\kappa \simeq 1.46$), but adding a single rank-5 radial-envelope nuisance drives $\kappa \simeq 35.5$ with a column overlap 0.996. Thus full rank within a chosen basis is not evidence that the basis is complete. Applying this framework to MnF_2 reconciles the lowest-order rank-3 O_{32}^- octupolar description with the recent polarized-neutron reconstruction of an important symmetry-allowed rank-5 O_{52} component, while keeping polarized-INS chirality and elastocaloric response in their proper tensor classes. The resulting rule is simple: same symmetry does not imply same tensor, same rank, proportional amplitude, or model-independent identifiability.

I. INTRODUCTION

The experimental language of altermagnetism now spans several mathematically distinct objects: Néel order, ferroic magnetic multipoles, piezomagnetic and thermodynamic response tensors, nonrelativistic electronic spin splitting, magnon chirality, and symmetry-selective Hamiltonian parameters.[1–3] These objects can carry the same symmetry character and reverse under the same magnetic-domain operation while remaining different expectation values or response functions. Treating them as interchangeable obscures what a particular experiment has actually established.

MnF_2 is an unusually sharp test case. Bhowal and Spaldin identified the ferroic magnetic octupole O_{32}^- , with real-space form $xy m_z$, as the lowest-order ferroically ordered magnetic quantity natural to the d -wave altermagnetic state.[4] Polarized-neutron theory and experiment established domain-sensitive magnon chirality.[5, 6] Resonant-x-ray and neutron diffraction calculations predict static multipolar signatures.[7] A 2026 polarized-neutron-diffraction (PND) preprint reconstructs the real-space magnetization density, resolving ligand spin polarization and an important symmetry-allowed rank-5

O_{52} contribution.[8] Independently, elastocaloric measurements under symmetry-matched strain and field provide a bulk thermodynamic probe of the altermagnetic transition.[9]

These results are complementary, not contradictory. A polarization-odd inelastic-neutron signal is not a measurement of a unique static multipole rank. A fitted exchange difference is not itself an order parameter. A thermodynamic derivative that transforms like an altermagnetic multipole is not numerically equal to a real-space multipole moment. Conversely, observing a substantial rank-5 component does not invalidate a rank-3 order-parameter description when both occupy the same symmetry sector.

The purpose of this paper is therefore narrower and more operational: to state which tensor each major probe constrains and to specify when a multipole rank is identifiable. The central principle is

$$\begin{aligned} \text{same symmetry sector} &\not\Rightarrow \text{same tensor,} \\ \text{same tensor} &\not\Rightarrow \text{same rank,} \\ \text{same rank} &\not\Rightarrow \text{same amplitude.} \end{aligned} \tag{1}$$

This paper occupies the tensor/rank-identifiability layer of a broader symmetry-selective observability program. It is self-contained: no conclusion requires a particular microscopic exchange model, and no dynamical measurement is used as a direct row in a static multipole inversion

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unless a model explicitly supplies that map.

II. OBJECT HIERARCHY AND MEASUREMENT MAPS

Let the microscopic Hamiltonian be

$$H = H_0 + \sum_a \lambda_a V_a, \quad (2)$$

where λ_a denotes exchange, dipolar, anisotropy, magnetoelectric, or other microscopic coordinates. The coordinates determine the state and excitations, but they are not themselves measured order parameters.

The equilibrium magnetization density is the one-point expectation value

$$\mathbf{m}(\mathbf{r}) = \langle \hat{\mathbf{m}}(\mathbf{r}) \rangle. \quad (3)$$

Static magnetic multipoles are symmetry-adapted moments of this density. A Cartesian rank-3 example is

$$O_{ijk} = \int d^3r r_i r_j m_k(\mathbf{r}), \quad (4)$$

with an equivalent spherical/tesseral decomposition into O_{KQ} .^[4]

A second static object class consists of response tensors. For example, a piezomagnetic coefficient may be defined schematically by

$$\Lambda_{i,jk} = \left. \frac{\partial M_i}{\partial \sigma_{jk}} \right|_{\sigma=0}, \quad (5)$$

where stress is time-even and magnetization is time-odd, so the appropriate Λ component reverses between time-reversed magnetic domains. The tensor Λ may be symmetry-linked to a multipolar order parameter, but it is a derivative of a response, not the multipole moment itself. The same logic applies to strain-field thermodynamic observables: a free-energy coupling of the form

$$f_{\text{int}} = -g_\Gamma \Phi_\Gamma X_\Gamma, \quad X_\Gamma \sim H_i \epsilon_\mu, \quad (6)$$

can establish the symmetry of an order parameter Φ_Γ without uniquely determining which real-space rank dominates its microscopic expansion.^[1, 2, 9]

The dynamical magnetic response is instead a two-point object. Define

$$S_{\alpha\beta}(\mathbf{Q}, \omega) = \int_{-\infty}^{\infty} dt e^{i\omega t} \langle \hat{M}_\alpha(\mathbf{Q}, t) \hat{M}_\beta(-\mathbf{Q}, 0) \rangle. \quad (7)$$

Writing $\hat{M} = \langle \hat{M} \rangle + \delta \hat{M}$ gives the exact separation

$$S_{\alpha\beta} = 2\pi \langle M_\alpha(\mathbf{Q}) \rangle \langle M_\beta(-\mathbf{Q}) \rangle \delta(\omega) + S_{\alpha\beta}^c(\mathbf{Q}, \omega). \quad (8)$$

The first term is the disconnected one-point contribution and the second is the connected correlation tensor. Interpreting the second term as purely inelastic requires a

specified broken-symmetry/domain state with no unre-solved static configurational mixture. This qualification matters for macroscopic antiferromagnetic domains. If two time-reversed domains have weights $p_\pm$ and Fourier magnetizations $M_\pm = \pm M_0$, then

$$\overline{M} = (p_+ - p_-)M_0 \equiv \eta_D M_0, \quad (9)$$

while an incoherent domain average of the magnetic intensity is

$$|\overline{M}|^2 = p_+ |M_0|^2 + p_- |M_0|^2 = |M_0|^2. \quad (10)$$

Thus an equal-domain ensemble can have $\overline{M} = 0$ while retaining finite magnetic diffraction. In the correlation-function language the missing one-point weight is carried by a static connected variance proportional to $(1 - \eta_D^2)|M_0|^2$, appearing at zero or quasi-zero frequency according to domain dynamics and experimental resolution. By contrast, the signed nuclear-magnetic interference term in polarized diffraction is linear in the domain-averaged amplitude and therefore scales with η_D . Equations (9)–(10) are also why one-point/two-point statements in this paper are always understood relative to a declared domain state and averaging procedure.

Finally, experiments return neither microscopic coordinates nor these tensors directly. For probe p ,

$$y_p = \mathcal{T}_p[\mathcal{P}_p(\mathcal{X})] + n_p, \quad (11)$$

where $\mathcal{X}$ denotes the relevant tensor object, $\mathcal{P}_p$ the probe kernel, $\mathcal{T}_p$ the instrumental transfer operation, and n_p statistical and systematic error. Figure 1 summarizes the hierarchy.

III. STATIC MULTIPOLES ARE NOT DYNAMICAL CHIRALITY

A. Elastic one-point sector

Magnetic neutron diffraction is governed by the transverse Fourier magnetization

$$\mathbf{M}_\perp(\mathbf{Q}) = \hat{\mathbf{Q}} \times [\mathbf{M}(\mathbf{Q}) \times \hat{\mathbf{Q}}], \quad (12)$$

$$\mathbf{M}(\mathbf{Q}) = \int d^3r e^{i\mathbf{Q} \cdot \mathbf{r}} \mathbf{m}(\mathbf{r}). \quad (13)$$

In PND the nuclear-magnetic interference term provides signed sensitivity to the magnetic structure factor. For MnF_2 , the 2026 PND analysis writes the relevant intensities schematically as^[8]

$$I^\pm = I_n + I_m \pm I_{\text{nm}}, \quad (14)$$

so polarization reversal isolates the interference contribution. A sufficiently diverse reflection set therefore constrains the spatial structure of $\mathbf{m}(\mathbf{r})$ and its multipole content.

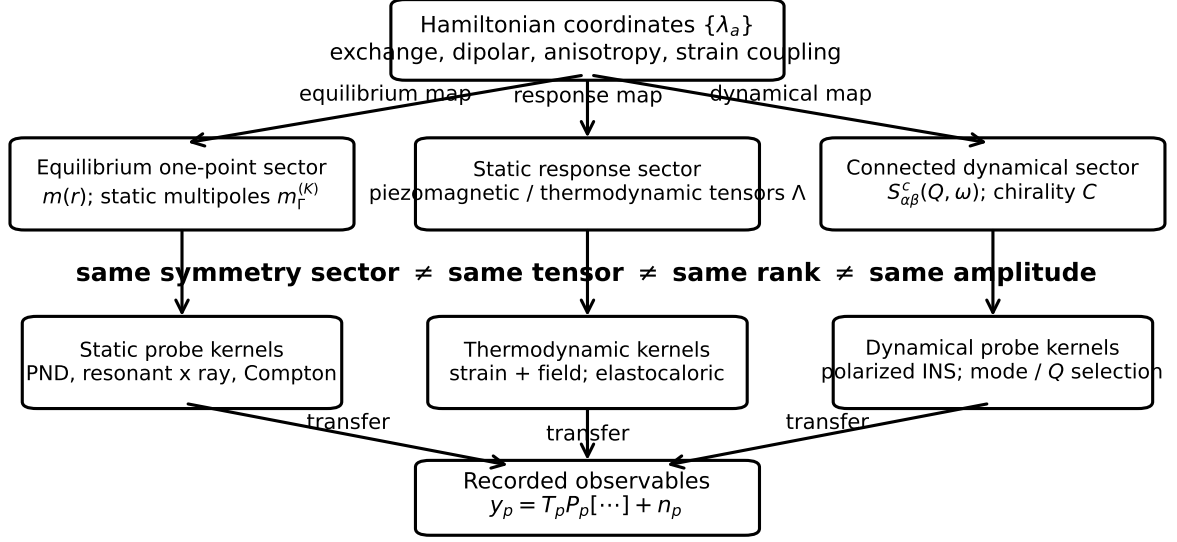


FIG. 1. Probe–tensor correspondence. Hamiltonian coordinates generate equilibrium, static-response, and dynamical sectors, after which each experiment applies a probe-specific projection and instrumental transfer map. Objects can share a symmetry character without being identical tensors, ranks, or amplitudes. The diagram is schematic, not a fitted model. Plotting code was prepared with AI assistance and checked by the author against the manuscript definitions.

B. Connected dynamical sector

Polarized inelastic neutron scattering (INS) accesses $S_{\alpha\beta}^c$ rather than the static density. Its antisymmetric component can be represented by

$$\mathcal{C}_\gamma(\mathbf{Q}, \omega) = \frac{i}{2} \epsilon_{\gamma\alpha\beta} [S_{\alpha\beta}^c(\mathbf{Q}, \omega) - S_{\beta\alpha}^c(\mathbf{Q}, \omega)]. \quad (15)$$

Within the Blume–Maleev formalism, the incident-polarization-odd magnetic intensity contains a term proportional to $\mathbf{P} \cdot \mathbf{C}$ after the usual transverse and geometry factors.[13–15] Schematically,

$$I(+\mathbf{P}) - I(-\mathbf{P}) \propto 2\mathbf{P} \cdot \mathbf{C}, \quad (16)$$

followed by instrumental convolution. This is the object accessed by the domain-dependent chiral magnon measurements in MnF_2 . [6]

The distinction survives energy integration. The connected sum rule is

$$\int \frac{d\omega}{2\pi} S_{\alpha\beta}^c(\mathbf{Q}, \omega) = \langle \delta M_\alpha(\mathbf{Q}) \delta M_\beta(-\mathbf{Q}) \rangle. \quad (17)$$

The right-hand side is an equal-time *two-point fluctuation correlator*, not the one-point multipole in Eq. (4). Thus even perfect energy integration of the inelastic sector does not convert magnon chirality into a unique static multipole rank.

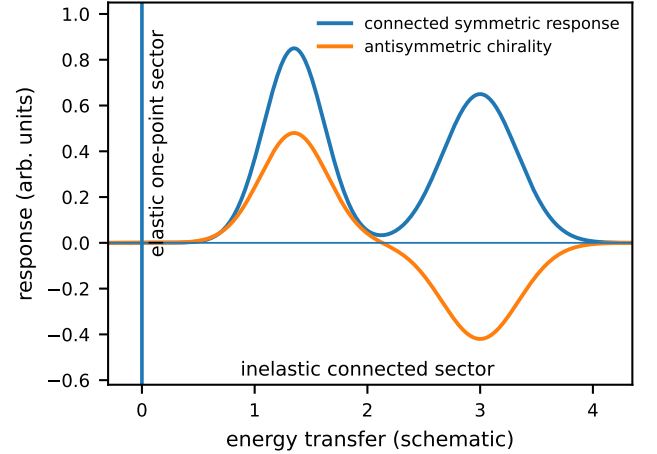


FIG. 2. Schematic separation of the elastic one-point contribution from the connected inelastic sector. The symmetric connected response and antisymmetric chiral response can occupy the same excitation spectrum while remaining distinct from the $\omega = 0$ equilibrium one-point contribution. The curves are schematic, not MnF_2 data. Plotting code was prepared with AI assistance and checked against Eqs. (8)–(17).

IV. SYMMETRY SECTOR VERSUS MULTIPOLE RANK

Let Γ denote a symmetry sector of the ordered state. The static multipole space carrying that symmetry can

contain several ranks,

$$\mathcal{V}_\Gamma = \bigoplus_{K \in \mathcal{K}_\Gamma} \mathcal{V}_\Gamma^{(K)}. \quad (18)$$

A selection rule can therefore identify Γ without uniquely identifying K . This distinction is increasingly important because first-principles work finds that several local magnetic multipoles can contribute comparably to alternating magnetic observables.[10, 11]

For MnF_2 , the rank-3 $O_{32} \propto xyz^2 m_z$ multipole is the lowest-order ferroic magnetic multipole associated with the d -wave alternating magnetic symmetry.[4] The recent PND analysis emphasizes that a higher-order O_{52} component with real-space structure proportional to $xyz^2 m_z$ is also symmetry allowed and is important in the reconstructed Mn spin-density asphericity.[8] These statements are compatible because “lowest allowed order parameter” and “largest experimentally reconstructed aspherical basis component” are different claims.

A. Local identifiability

Collect the candidate static multipole amplitudes in the selected sector into

$$\mathbf{m}_\Gamma = (m_1, \dots, m_n)^T. \quad (19)$$

After linearization about nuisance/background parameters, measurements at settings $r = 1, \dots, R$ may be written

$$\mathbf{y} = K_\Gamma \mathbf{m}_\Gamma + \mathbf{n}, \quad (20)$$

with covariance Σ . The whitened kernel is

$$\tilde{K}_\Gamma = \Sigma^{-1/2} K_\Gamma, \quad (21)$$

and the corresponding local Fisher matrix is

$$F_\Gamma = K_\Gamma^T \Sigma^{-1} K_\Gamma = \tilde{K}_\Gamma^T \tilde{K}_\Gamma. \quad (22)$$

All n candidate components are locally identifiable only if

$$\text{rank}(F_\Gamma) = \text{rank}(K_\Gamma) = n. \quad (23)$$

Practical identifiability additionally requires that the smallest singular value of $\tilde{K}_\Gamma$ be large enough relative to uncertainty; a formally full-rank but ill-conditioned kernel can still make rank separation experimentally unstable.

B. Conditional identifiability: nuisance space, basis completeness, and origin

Equation (23) is a statement inside a declared model. Let N collect nuisance directions such as ligand moments, radial/form-factor parameters, extinction/background

terms, or additional symmetry-allowed multipoles. In whitened coordinates define

$$Q_N = \tilde{N}(\tilde{N}^T \tilde{N})^+ \tilde{N}^T, \quad (24)$$

where $+$ denotes the Moore–Penrose inverse. The target information that remains after allowing the nuisance space is carried by

$$\tilde{K}_{\Gamma, \text{eff}} = (I - Q_N) \tilde{K}_\Gamma, \quad (25)$$

with conditional Fisher matrix

$$F_{\Gamma|N} = \tilde{K}_{\Gamma, \text{eff}}^T \tilde{K}_{\Gamma, \text{eff}}. \quad (26)$$

A rank claim is locally identifiable only if $\text{rank}(F_{\Gamma|N}) = n$ with adequate conditioning. Full rank of K_Γ alone is therefore insufficient when an omitted allowed direction can mimic a target column.

The basis itself must also be declared. Magnetic multipole coefficients are defined relative to an origin, local axes, radial basis, and truncation. Translations can mix Cartesian moments of different order when lower moments are nonzero, and different radial envelopes can produce nearly collinear diffraction signatures over a finite Q range. For MnF_2 we therefore use the Mn-site-centered, crystallographic-axis convention employed in the cited multipole reconstruction when discussing $P_{2,-2}$ and $P_{4,-2}$; the rank statement is conditional on that convention rather than an origin-free scalar fact.

This gives an immediate experimental-design rule. Symmetry determines which columns are allowed, while the probe settings and nuisance model determine whether the target columns remain independent after projection. A unique rank claim therefore requires more than observing a symmetry-allowed reflection or obtaining full rank in a reduced fit.

C. Published-geometry MnF_2 kernel stress test

The public PND report gives enough information for a limited but useful conditioning test: $a = b = 4.87 \text{ \AA}$, $c = 3.30 \text{ \AA}$, $x_F = 0.3049$, a cutoff $\sin \theta / \lambda < 0.8 \text{ \AA}^{-1}$, 57 threshold-selected even reflections in the 1 T multipole refinement, and the site-symmetry basis Y_{20} , $Y_{2,-2}$, Y_{40} , and $Y_{4,-2}$. [8] The full selected hkl list, experimental covariance, extinction covariance, and Slater-function covariance are not tabulated, so the following calculation is explicitly a *published-geometry surrogate*, not a re-refinement of the data.

We enumerate reciprocal-lattice points within the reported cutoff, impose the even/odd parity structure of the rutile Mn and ligand contributions, and use dimensionless radial proxies $g_l(q) \propto q^l \exp[-\beta(q/q_{\text{max}})^2]$. From the even candidate pool we select 57 rows with the largest leverage in the reduced four-column basis $\{F P_{00}, F P_{20}, \text{Mn } P_{2,-2}, \text{Mn } P_{4,-2}\}$; arbitrary column

TABLE I. Conditioning of the published-geometry surrogate. Singular values refer to unit-norm columns; $\kappa = \sigma_{\max}/\sigma_{\min}$. The last column is the largest absolute cosine overlap between distinct normalized columns. These numbers diagnose geometric/model collinearity only and are not experimental parameter uncertainties.

Model	$\sigma_{\min}$	κ	max overlap
Even, 4-column base	0.805	1.46	0.304
Base + rank-7 angular	0.752	1.62	0.304
Base + rank-5 radial	0.0418	35.5	0.996
Base + rank-5/F radial	0.0382	40.5	0.996
Full 5-column count match	0.844	1.35	0.254

scales are removed by unit-norm normalization. A count-matched full surrogate additionally includes 140 low- Q odd rows to expose the spherical Mn P_{00} block. The exact construction and selected surrogate rows are supplied with the manuscript source.

The reduced basis is therefore geometrically capable of separating $P_{2,-2}$ from $P_{4,-2}$ in this surrogate: their normalized overlap is only 0.070, and the even-sector condition number is 1.46. Importantly, basis augmentation does not automatically destroy this result: adding a same-symmetry higher-rank $P_{6,-2}$ proxy changes the condition number only to 1.62. The result changes qualitatively for a different kind of model incompleteness. A single nuisance column corresponding to the derivative of the $P_{4,-2}$ radial envelope with respect to its decay parameter has overlap $|\widehat{k}_{4,-2}^T \widehat{k}_{4,-2}^{(r)}| = 0.996$ and drives the condition number to 35.5 while the matrix remains formally full rank. Adding a ligand radial-envelope nuisance raises it to 40.5. Varying the Mn radial-decay proxy over a broad dimensionless range ($\beta_{\text{Mn}} = 0.4\text{--}1.6$) leaves the rank-5/radial overlap above 0.994 and the condition number between approximately 31 and 37. This is an explicit counterexample to the inference “full rank $\Rightarrow$ complete rank identification,” while also showing that the failure is specific to a near-degenerate nuisance family rather than an inevitable consequence of adding parameters.”

The count-matched full surrogate also illustrates the usefulness of parity separation: the idealized spherical Mn P_{00} column is orthogonal to the even Mn $P_{2,-2}$ and $P_{4,-2}$ columns, while ligand terms couple to both sectors. This explains why odd reflections can calibrate the dominant spherical Mn contribution without, by themselves, establishing which even-sector anisotropic basis is complete.

The published refinements strongly favor a model containing Mn $P_{4,-2}$ over the tested lower-rank alternative for the selected even reflections, with reported $\chi^2/N_{\text{peak}} = 1.64$ versus 3.42 for the $P_{2,-2}$ model. [8] However, those tabulated models do not provide a simultaneous covariance matrix for $P_{2,-2}$, $P_{4,-2}$, ligand-shape, and radial nuisance directions. The strongest defensible statement is therefore that the data support an important rank-5 component within the tested reconstruction family;

unique rank closure requires the augmented-kernel test in Eqs. (25)–(26) or an equivalent full covariance analysis.

D. Complementary static probes as a rank lift

For statistically independent static probes p , the information matrices add,

$$F_{\text{joint}} = \sum_p K_p^T \Sigma_p^{-1} K_p. \quad (27)$$

PND, resonant x-ray diffraction, and momentum-density measurements can therefore lift a rank degeneracy if their kernels weight candidate basis functions differently. By contrast, polarized INS does not automatically add another row to Eq. (20), because its primary object is S^c , not $\mathbf{m}_\Gamma$.

Dynamic data can constrain the same *microscopic model parameters* only after a model supplies the relevant Jacobian. If θ is a shared model parameter vector and probe p predicts mean data $\mu_p(\theta)$, then

$$F_\theta = \sum_p J_p^T \Sigma_p^{-1} J_p, \quad J_p = \frac{\partial \mu_p}{\partial \theta}. \quad (28)$$

Equation (28), not symmetry alone, is the appropriate route to quantitative amplitude closure between static and dynamical measurements.

V. PROBE-TENSOR DICTIONARY FOR MNF₂

Table II summarizes the main probe classes. “Direct” means direct with respect to the listed mathematical object after the standard probe kernel is modeled; it does not mean model-free inversion into every other object in the hierarchy.

A. Static real-space density: polarized neutron diffraction

The 2026 PND preprint exploits symmetry-selective reflections to separate the dominant spherical Mn²⁺ contribution from weak Mn asphericity and ligand spin density.[8] It reports a fluorine moment near 0.028 μ_B at 1 T and reconstructs an anisotropic Mn density consistent with an important rank-5 O_{52} contribution. The published model comparison strongly improves the even-reflection fit when Mn $P_{4,-2}$ is included, but the reported table compares alternative finite bases rather than presenting the simultaneous target–nuisance covariance required for model-independent rank uniqueness. Because the inversion uses many reflections and an explicit site-centered density model, the correct statement is about the static real-space magnetization density within that reconstruction family, not about an inelastic chiral correlator or an origin-free unique rank.

TABLE II. Probe–tensor dictionary for the MnF₂ program.

Probe	Primary mathematical object / selection	Can establish most directly	Does not establish by itself
Unpolarized neutron diffraction	$ \mathbf{M}_\perp(\mathbf{Q}) ^2$; reflection symmetry	static magnetic structure	signed multipole phase or unique rank
Polarized neutron diffraction	$\mathbf{M}_\perp(\mathbf{Q})$; interference, domain, reflection	static magnetization density; rank with sufficient $\mathbf{Q}$ coverage	magnon chirality
Polarized INS	$S_{\alpha\beta}^z$ and $\mathcal{C}$; polarization, mode, momentum	dynamical chirality and mode correlations	unique static multipole rank
Resonant x-ray diffraction	resonance-weighted multipoles; polarization/helicity, azimuth	symmetry- and rank-selective static amplitudes	unique microscopic exchange parameter
Magnetic Compton / momentum density	spin-resolved momentum density; direction, domain	momentum-density signature of ferroic multipolar order	real-space multipole rank without a model
Elastocaloric / strain-field thermodynamics	free-energy derivatives in symmetry-matched field-strain channel	bulk thermodynamic symmetry of order and critical response	unique local multipole rank or magnon chirality
Spin-resolved band spectroscopy	one-particle spectral function; momentum, spin, domain	momentum-space spin splitting	local static multipole amplitude

B. Static tensor selection: resonant x rays and momentum density

Lovesey derives MnF₂ Bragg signatures in resonant x-ray scattering whose polarization, helicity, and azimuthal dependence select axial magnetic multipoles.[7] Such energy- and polarization-dependent kernels are valuable precisely because they differ from neutron form factors and can improve Eq. (27). Bhowal and Spaldin predict a nonzero magnetic Compton signature associated with ferroic octupolar order despite zero net magnetization,[4] while momentum-density theory provides a broader route to diagnosing altermagnetic spin asymmetry.[12]

C. Thermodynamic response: strain and field

Ohlendorf *et al.* combine symmetry-guided strain and magnetic-field tuning with elastocaloric measurements and first-principles theory to establish a bulk thermodynamic signature of the finite-temperature altermagnetic transition in MnF₂. [9] This is a particularly important cross-check because it does not rely on the same scattering kernel as PND or INS. At the same time, it should be interpreted as a thermodynamic response in the symmetry-matched field–strain channel. By itself it does not decide whether O_{32}^- , O_{52} , or a multicomponent expansion best represents the microscopic magnetization density.

D. Dynamical response: polarized INS

Polarized INS measures the antisymmetric connected dynamical response in Eq. (15). The observed polarization-odd spectral weight and its domain/mode reversals establish dynamical chirality.[6] They do not constitute a direct measurement of O_{32}^- or O_{52} . A microscopic Hamiltonian can relate the static and dynamic sectors, but the proportionality is model dependent.

VI. MNF₂: A HIERARCHY OF CLAIMS

The current literature supports the following hierarchy without tensor conflation.

(1) Symmetry and lowest-order multipolar language. The d -wave altermagnetic state admits a ferroic magnetic octupole as the lowest-order magnetic multipole

natural to the symmetry, and tensor/Landau descriptions connect that symmetry to bulk responses such as piezomagnetism.[1, 3, 4]

(2) Static microscopic density. PND reconstructs the equilibrium magnetization density and reports finite ligand polarization plus an aspherical Mn contribution in which rank-5 O_{52} structure is important.[8] This enriches rather than contradicts the lowest-order octupolar description.

(3) Thermodynamic order. Strain–field elastocaloric measurements establish a bulk symmetry-sensitive thermodynamic response associated with the altermagnetic transition.[9] The measured object is a free-energy derivative, not a local multipole coefficient.

(4) Dynamical chirality. Polarized INS establishes a domain-dependent antisymmetric connected magnon response.[6] This is a two-point dynamical observable.

(5) Electronic and microscopic quantities. Momentum-space spin splitting and exchange/dipolar/anisotropy parameters belong to still different layers. Recent first-principles work shows that quantitative spin splitting can receive comparable contributions from more than one multipole, reinforcing a multicomponent rather than single-rank interpretation.[10]

The phrase “direct observation” is therefore meaningful only relative to a specified object. PND can directly constrain the static magnetization density; polarized INS can directly constrain dynamical chirality; elastocaloric measurements can directly constrain a thermodynamic response. None of these statements, by itself, licenses direct inversion into all other layers.

VII. CROSS-PROBE CLOSURE ON THE SAME DOMAIN STATE

A strong correspondence test should use the same crystal and reproducibly trained magnetic domain. Let $d = \pm 1$ label time-reversed domains. For a time-odd static multipole amplitude m_a and an appropriate time-odd static response tensor component Λ ,

$$m_a(-d) = -m_a(d), \quad \Lambda(-d) = -\Lambda(d). \quad (29)$$

For the matched inelastic geometry, the chiral response reverses as

$$\mathcal{C}_\gamma(\mathbf{Q}, \omega; -d) = -\mathcal{C}_\gamma(\mathbf{Q}, \omega; d), \quad (30)$$

with mode and momentum conventions held fixed. These are symmetry/domain statements; they do not impose a common amplitude scale.

A same-crystal closure experiment should therefore proceed in three nested levels:

1. **Symmetry closure:** train both domains and verify the required sign reversal in static PND, thermodynamic response, and polarized INS after correcting for domain fraction.
2. **Rank closure:** predeclare the static basis, origin/local-axis convention, radial/form-factor family, and nuisance columns; choose PND and complementary static-probe settings to maximize the smallest singular value of the nuisance-projected joint kernel; then repeat the inference under allowed basis augmentation. Test whether O_{32}^- and O_{52} remain separable rather than merely fitted in one reduced model.
3. **Microscopic amplitude closure:** only after the first two steps, fit all datasets through a Hamiltonian/electronic model that predicts the static density, thermodynamic response, and $S^c(\mathbf{Q}, \omega)$. Require one parameter set to reproduce amplitudes and signs within uncertainty.

A model can pass symmetry closure and fail amplitude closure. That failure is scientifically useful: it localizes the missing physics to the constitutive map between tensors rather than to their transformation law.

Temperature provides an additional discriminant. Static multipoles and thermodynamic susceptibilities may have Landau-controlled scaling near a continuous transition, while mode-resolved chiral spectral weight also depends on excitation energies, damping, occupation, and experimental integration windows.[1, 2] Equal critical exponents or strict amplitude proportionality should therefore be derived, not assumed.

VIII. FALSIFICATION RULES AND SCOPE

The correspondence framework gives eight practical rules.

1. **Object rule.** State whether the inferred quantity is a Hamiltonian coordinate, equilibrium one-point tensor, static response tensor, connected dynamical correlator, one-particle spectrum, or recorded intensity.
2. **State/ensemble rule.** Declare the broken-symmetry/domain state and averaging procedure. A vanishing ensemble one-point moment does not imply vanishing diffraction intensity, and unresolved domain variance can contribute at zero or quasi-zero frequency.

3. **Basis/origin rule.** Declare the multipole origin, local axes, radial basis, truncation, ligand degrees of freedom, and other symmetry-allowed components before making a rank claim.

4. **Symmetry rule.** A selection rule establishes a symmetry sector, not automatically a unique multipole rank.

5. **Conditional-rank rule.** Claim rank separation only when the nuisance-projected kernel in Eq. (25) has full rank with adequate conditioning and the result is stable to physically allowed basis augmentation. Full rank in a reduced basis is not sufficient.

6. **Integration rule.** Energy integration of an inelastic connected response yields an equal-time two-point correlator, not a unique static one-point multipole.

7. **Domain-parity rule.** Time-odd static and dynamical objects must show the appropriate domain reversal after domain-population and geometry corrections; sign-sensitive interference and sign-insensitive intensity need not have the same domain dilution.

8. **Model rule.** Quantitative relations among multipole amplitudes, thermodynamic tensors, magnon chirality, band splitting, and microscopic interactions are constitutive/model statements. Closure within one model class is not proof that the model class is complete; independent probes should actively test nuisance and omitted-basis alternatives.

This paper deliberately does not claim that one rank universally dominates all alternating observables, that a chiral magnon spectrum uniquely determines a static multipole, or that one microscopic exchange coordinate controls every static and dynamic signature. Nor does the surrogate calculation reproduce the experimental PND covariance. It is a controlled counterexample demonstrating why basis completeness and conditioning are logically prior to unique rank closure.

IX. CONCLUSION

MnF₂ now supports several experimentally grounded descriptions simultaneously: a lowest-order ferroic octupolar order-parameter language, reconstructed higher-rank structure in the real-space magnetization density, a bulk strain-field thermodynamic response, dynamical magnon chirality, momentum-space spin structure, and weak microscopic Hamiltonian coordinates. The correct synthesis is a correspondence map, not an identity claim.

Three results capture the hardened statement. Equation (8) separates one-point order from connected correlations while Eqs. (9)–(10) show why this separation must be interpreted in a declared domain ensemble. Equation (25) makes rank identifiability conditional on the nuisance

space rather than on the target basis alone. Finally, Table I gives an explicit counterexample: a reduced MnF_2 PND geometry surrogate is well conditioned, yet adding one physically plausible radial-shape nuisance leaves formal full rank while degrading the condition number by more than an order of magnitude.

These results explain why rank-3 O_{32}^- and rank-5 O_{52} descriptions can coexist in MnF_2 , why the recent PND reconstruction can support an important O_{52} component without proving an origin- and basis-independent unique rank, and why polarized-INS chirality remains a distinct dynamical observable. The practical discipline is therefore: define the object and state, predeclare the basis and nuisance space, write the probe kernel, project out nuisance directions, test conditioning and basis augmentation, verify domain parity, and only then infer microscopic correspondence. This converts the apparent competition among “multipole,” “chirality,” “piezomagnetic,” and “spin-splitting” descriptions into a falsifiable inverse problem rather than a competition of labels.

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DATA AVAILABILITY

No new experimental data were generated for this work. All experimental numerical values discussed are drawn from the cited literature. The published-geometry surrogate in Table I is a new calculation based only on publicly reported MnF_2 lattice parameters, cutoff, parity rules, reflection counts, and symmetry-allowed basis functions; it is not a re-refinement of experimental data. The exact Python script, selected surrogate reflection tables, normalized-kernel summary, manuscript source, and schematic-figure sources are included in the accompanying preparation package.

COMPETING INTERESTS

The author declares no competing interests.

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